Zhuoheng Ying

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Education

Tongji University *Shanghai, China*
M.S. in Transportation Engineering *Sep. 2025 - Jun. 2028*

- Selected coursework: Applied Statistics, Traffic Data Analysis and Applications, Transportation System Control, and Traffic Flow Theory and Simulation.

Chang'an University *Xi'an, China*
B.S. in Transportation Engineering *Sep. 2021 - Jun. 2025*

- Rank: 3/117; GPA: 3.65/5.0; Outstanding Graduate (2025). Core coursework included transportation planning, operations research, traffic psychology and ergonomics, GIS, and urban traffic management.

University College Dublin *Dublin, Ireland*
International Study Abroad Program (ISA) *Jul. 2023 - Aug. 2023*

Employment

Zhijia Xincheng (Shanghai) Intelligent Technology Co., Ltd. *Shanghai, China*
Intelligent Driving Product & Testing Intern | Horizon Robotics x AUMOVIO *Apr. 2026 - Jul. 2026*

- Analyzed C-NCAP, US NCAP, i-VISTA, C-ICAP, C-IASI, and UNECE R79/R152 requirements; translated test scenarios, metrics, and pass/fail criteria into structured regulatory analyses and test requirement documents.
- Mapped regulatory clauses to AEB, LKA, and LCC functions; developed compliance checklists, test cases, and requirements traceability matrices; supported product reviews and issue-closure tracking.
- Developed a prototype LLM-based regulatory analysis agent for document retrieval, clause-level Q&A, cross-standard comparison, and structured information extraction.
- Participated in the Luna 6 urban/highway on-road review and a four-vehicle benchmark study, evaluating functionality, safety, stability, comfort, system disengagements, takeover behavior, and user experience.

Selected Publications

[P3] eHMI Design Effects on Pedestrian Behavior at Unsignalized Intersections.
Fourth International Conference on Frontiers of Traffic and Transportation Engineering (FTTE 2025). Accepted for presentation and publication. Third author.

[P2] Factors Influencing Commuter Travel Mode Choice: A Study of Travel Survey Data in the Puget Sound Region of the Northwestern United States.
Third International Conference on Digital Economy and Computer Application (DECA 2023). EI-indexed. Second author.

[P1] Traffic Flow Prediction Based on T-GCN in Extreme Weather: A Case Study of Beijing.
CONF-MSS 2023. Co-first author.

Research Focus

Core interests: ADAS product definition and verification, regulatory intelligence, pedestrian-automated vehicle interaction, multimodal human-factors experimentation, computer vision, and traffic-state prediction.

Research question: How can data-driven models and human-machine interfaces improve the safety, interpretability, and user acceptance of automated mobility in complex urban environments?

Research Experience

Pedestrian-Automated Vehicle Interaction at Unsignalized Intersections *Shanghai, China*
Research Contributor | Third Author *Mar. 2025 - Oct. 2025*

- Investigated how external human-machine interface (eHMI) designs affect pedestrian crossing decisions, cognitive responses, and perceived safety at unsignalized intersections.
- Contributed to a Unity/VR experimental platform and multimodal data collection covering behavioral, physiological, video, and questionnaire measures.
- Led data cleaning and integration, statistical modeling, result visualization, and English manuscript preparation and revision.

Research Experience (continued)

YOLO-Based Road Object Detection Project *Shanghai, China*
Project Lead *Oct. 2025 - Feb. 2026*

- Built a YOLOv11 detection pipeline for pedestrian and vehicle recognition using public datasets; completed annotation and COCO/YOLO format conversion.
- Executed training, validation, and inference; evaluated Precision, Recall, and mAP; implemented OpenCV-based bounding-box and class-label visualization.

Traffic Flow Forecasting under Extreme Weather *Xi'an, China*
Co-First Author *Dec. 2022 - Sep. 2023*

- Developed a T-GCN model combining graph convolutional networks and GRU units to capture road-network spatial dependence and traffic-flow temporal dynamics using Beijing traffic and weather data.
- Conducted data preprocessing, benchmark experiments, result analysis, and manuscript writing and revision.

Commuter Travel Mode Choice Factors *Xi'an, China*
Second Author *Sep. 2023 - Dec. 2023*

- Analyzed Puget Sound regional travel-survey data and contributed to data cleaning, variable construction, model analysis, and interpretation.

Honors and Awards (Selected)

2025 Third Prize, 22nd “Huawei Cup” China Post-Graduate Mathematical Contest in Modeling.
2025 Outstanding Graduate, Chang'an University.
2023 Honorable Mention, Mathematical Contest in Modeling (top 7%).
2023 Second Prize, Shaanxi Division, China Undergraduate Mathematical Contest in Modeling.
2023 Third Prize, 14th Shaanxi Undergraduate Mathematics Competition (Non-Mathematics Category).

Technical Skills

Programming: Python; working knowledge of C++ and SQL; PyTorch, Hugging Face Transformers, MS-Swift, OpenCV, NumPy, and Flask.
AI / ML: YOLO, Transformer, CLIP, BLIP-2, Qwen, LLaVA, T-GCN, GRU, reinforcement learning, model fine-tuning, and multimodal learning.
Simulation & Analysis: Unity, VR experimentation, CARLA, GPUDrive, statistical modeling, questionnaire analysis, model evaluation, and data visualization.
Transportation & GIS: ADAS regulations and test design; AEB, LKA, LCC; eHMI human-factors research; traffic-flow forecasting; ArcGIS spatial analysis.
Languages & Tools: Mandarin Chinese (native); English (IELTS 6.5, CET-6 528); Microsoft Office; LaTeX; PRC Class C1 Driver's License.

Leadership and Campus Engagement

Graduate Student Practice Department, Tongji University *Shanghai, China*
Staff Member *Sep. 2025 - Present*

- Plan campus practice activities and company visits; coordinate external partners, travel, venues, materials, on-site execution, and post-event reporting.

School Science and Technology Association, Chang'an University *Xi'an, China*
Staff Member *Sep. 2021 - Sep. 2023*

- Supported student innovation competitions through team and faculty coordination, logistics, documentation, on-site operations, and cross-department communication.